In [2]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [4]:
```
df = pd.read_csv("messy_student_data.csv")
df.head()
```

Out[4]:

	StudentID	Name	Age	Gender	City	Math_Score	Science_Score	Attendance_Percent	Enrolled_Date
0	S1112	Ananya Gupta	19	male	Salem	71.0	78.0	64.0	01/07/2025
1	S1102	Rahul Singh	18	F	Chennai	85.0	80.0	69.0	01/07/2025
2	S1077	Divya Pillai	22	male	Bangalore	96.0	88.0	85.0	15-06-2025
3	S1080	Rohan Verma	21	NaN	Bangalore	46.0	NaN	66.0	15-06-2025
4	S1041	aditi sharma	21	female	chennai	NaN	40.0	54.0	15-06-2025

In [5]:
```
print("Shape:", df.shape)
df.info()
```

```
Shape: (126, 9)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 126 entries, 0 to 125
Data columns (total 9 columns):
 #   Column              Non-Null Count  Dtype  
---  ------              --------------  -----  
 0   StudentID           126 non-null    object 
 1   Name                126 non-null    object 
 2   Age                 116 non-null    object 
 3   Gender              100 non-null    object 
 4   City                112 non-null    object 
 5   Math_Score          121 non-null    float64
 6   Science_Score       111 non-null    float64
 7   Attendance_Percent  123 non-null    float64
 8   Enrolled_Date       107 non-null    object 
dtypes: float64(3), object(6)
memory usage: 9.0+ KB
```

In [6]:
```
print(df.isnull().sum())
print("\nDuplicate rows:", df.duplicated().sum())
```

```
StudentID              0
Name                   0
Age                   10
Gender                26
City                  14
Math_Score             5
Science_Score         15
Attendance_Percent     3
Enrolled_Date         19
dtype: int64

Duplicate rows: 6
```

In [7]:
```
df = df.drop_duplicates().reset_index(drop=True)
df.shape
```

Out[7]: (120, 9)

In [8]:
```
df["Name"] = df["Name"].str.strip().str.title()
df["City"] = df["City"].str.strip().str.title()

gender_map = {"m": "M", "male": "M", "f": "F", "female": "F"}
df["Gender"] = df["Gender"].str.strip().str.lower().map(gender_map)
df["Gender"] = df["Gender"].fillna("Unknown")

df["City"] = df["City"].fillna("Unknown")
df[["Name", "City", "Gender"]].head()
```

Out[8]:

	Name	City	Gender
0	Ananya Gupta	Salem	M
1	Rahul Singh	Chennai	F
2	Divya Pillai	Bangalore	M
3	Rohan Verma	Bangalore	Unknown
4	Aditi Sharma	Chennai	F

In [9]:
```
df["Age"] = pd.to_numeric(df["Age"], errors="coerce")
df["Age"] = df["Age"].fillna(df["Age"].median()).astype(int)

for col in ["Math_Score", "Science_Score", "Attendance_Percent"]:
    df[col] = pd.to_numeric(df[col], errors="coerce")
    df[col] = df[col].clip(lower=0, upper=100)
    df[col] = df[col].fillna(df[col].median())

df[["Age", "Math_Score", "Science_Score", "Attendance_Percent"]].describe()
```

Out[9]:

	Age	Math_Score	Science_Score	Attendance_Percent
count	120.000000	120.00000	120.000000	120.000000
mean	19.866667	67.00000	68.150000	77.533333
std	1.269774	23.43057	17.998436	14.553831
min	18.000000	0.00000	35.000000	50.000000
25%	19.000000	51.00000	56.000000	66.750000
50%	20.000000	69.00000	66.000000	78.000000
75%	21.000000	86.00000	83.000000	90.250000
max	22.000000	100.00000	100.000000	100.000000

In [10]:
```
df["Enrolled_Date"] = pd.to_datetime(df["Enrolled_Date"], format="mixed", dayfirst=True, errors="coerce")
df["Enrolled_Date"] = df["Enrolled_Date"].fillna(df["Enrolled_Date"].mode()[0])
```

```
df["Enrolled_Date"].head()
```

Out[10]:

```
0   2025-07-01
1   2025-07-01
2   2025-06-15
3   2025-06-15
4   2025-06-15
Name: Enrolled_Date, dtype: datetime64[ns]
```

In [11]:

```
df.isnull().sum()
```

Out[11]:

```
StudentID            0
Name                 0
Age                  0
Gender               0
City                 0
Math_Score           0
Science_Score        0
Attendance_Percent   0
Enrolled_Date        0
dtype: int64
```

In [12]:

```
df["Total_Score"] = df["Math_Score"] + df["Science_Score"]
df["Average_Score"] = (df["Total_Score"] / 2).round(2)

def grade_from_avg(avg):
    if avg >= 90:
        return "A"
    elif avg >= 75:
        return "B"
    elif avg >= 60:
        return "C"
    elif avg >= 40:
        return "D"
    else:
        return "F"

df["Grade"] = df["Average_Score"].apply(grade_from_avg)
df["Pass_Fail"] = np.where(df["Average_Score"] >= 40, "Pass", "Fail")
df.head()
```

Out[12]:

	StudentID	Name	Age	Gender	City	Math_Score	Science_Score	Attendance_Percent	Enrolled_Date	Total_Score	A
0	S1112	Ananya Gupta	19	M	Salem	71.0	78.0	64.0	2025-07-01	149.0	
1	S1102	Rahul Singh	18	F	Chennai	85.0	80.0	69.0	2025-07-01	165.0	
2	S1077	Divya Pillai	22	M	Bangalore	96.0	88.0	85.0	2025-06-15	184.0	
3	S1080	Rohan Verma	21	Unknown	Bangalore	46.0	66.0	66.0	2025-06-15	112.0	
4	S1041	Aditi Sharma	21	F	Chennai	69.0	40.0	54.0	2025-06-15	109.0	

In [13]:

```
# Students with good attendance (>= 75%)
good_attendance = df[df["Attendance_Percent"] >= 75]
print(f"Students with attendance >= 75%: {len(good_attendance)} of {len(df)}")

# Top performers (average score >= 75)
top_performers = df[df["Average_Score"] >= 75].sort_values("Average_Score", ascending=False)
print(f"Top performers (avg >= 75): {len(top_performers)}")
top_performers[["StudentID", "Name", "Average_Score", "Grade"]].head(10)
```

```
Students with attendance >= 75%: 73 of 120
Top performers (avg >= 75): 42
```

Out[13]:

	StudentID	Name	Average_Score	Grade
42	S1085	Divya Pillai	98.0	A
11	S1053	Ananya Gupta	95.5	A
63	S1071	Rohan Verma	92.0	A
2	S1077	Divya Pillai	92.0	A
52	S1098	Meera Nair	91.5	A
24	S1044	Sneha Reddy	90.5	A
66	S1036	Ishita Bose	89.5	B
94	S1056	Karthik Iyer	89.5	B
16	S1066	Vikram Rao	89.0	B
73	S1002	Sneha Reddy	88.5	B

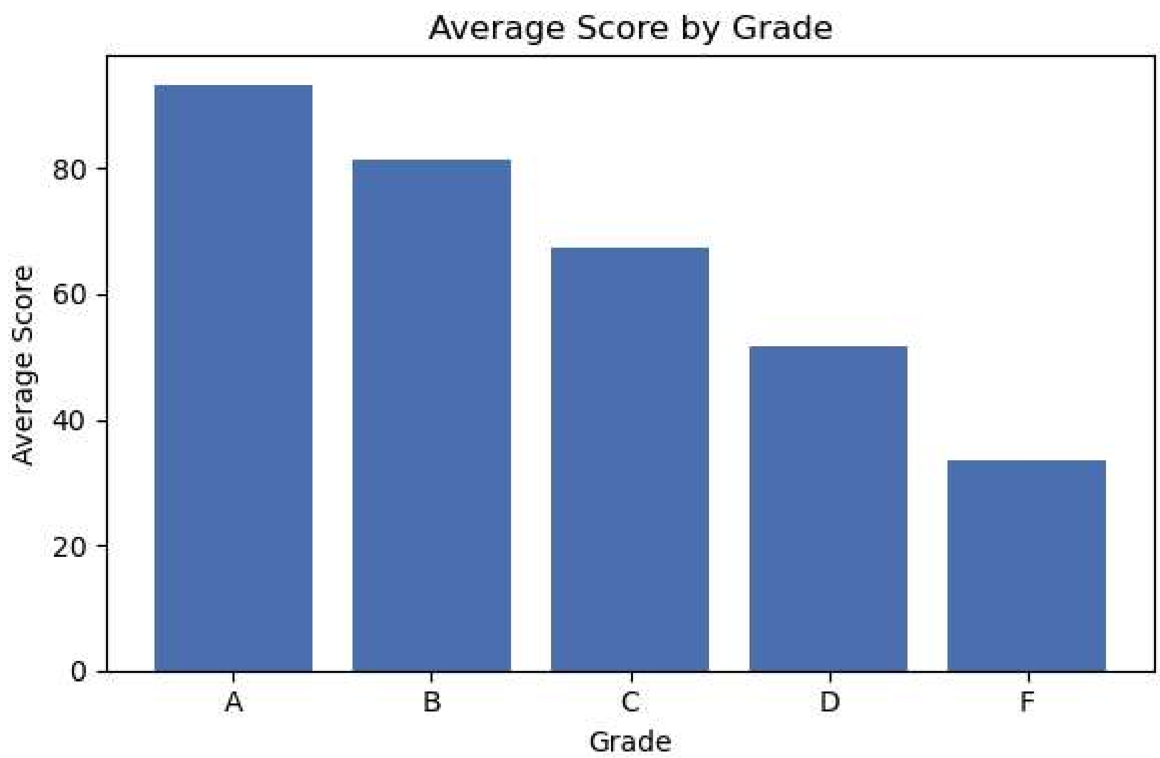
In [14]:

```
df.to_csv("cleaned_student_data.csv", index=False)
print("Saved cleaned_student_data.csv with shape", df.shape)
```

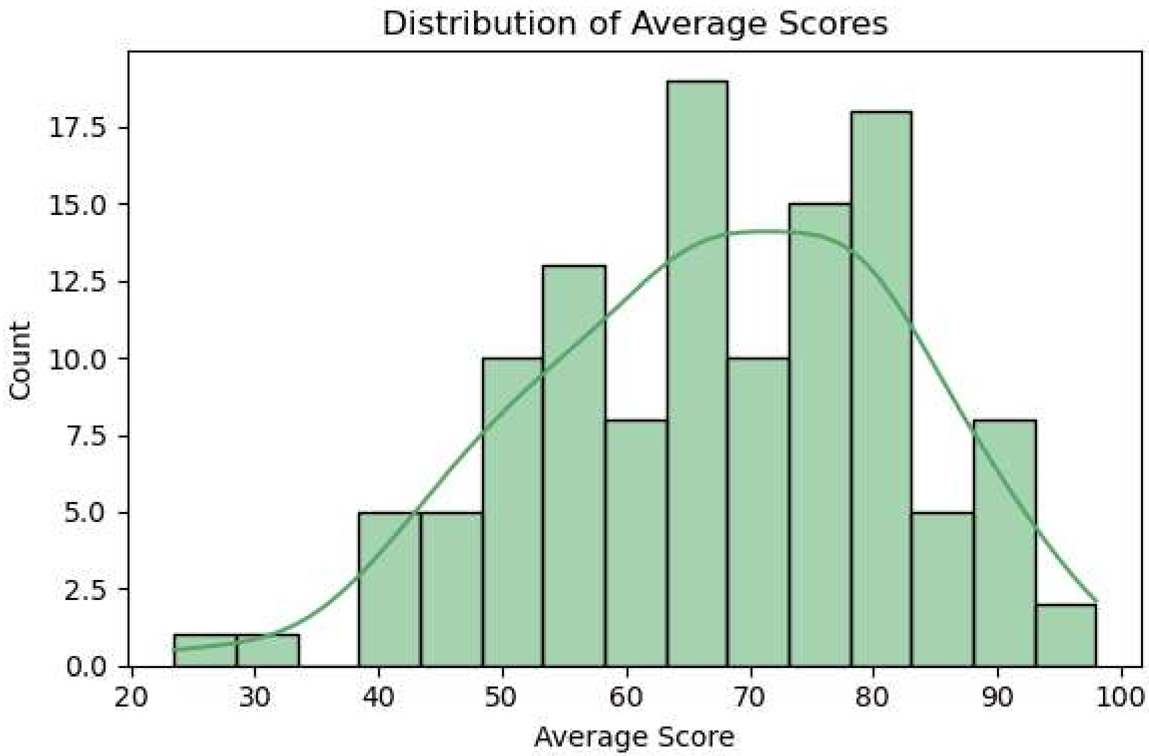
```
Saved cleaned_student_data.csv with shape (120, 13)
```

In [15]:

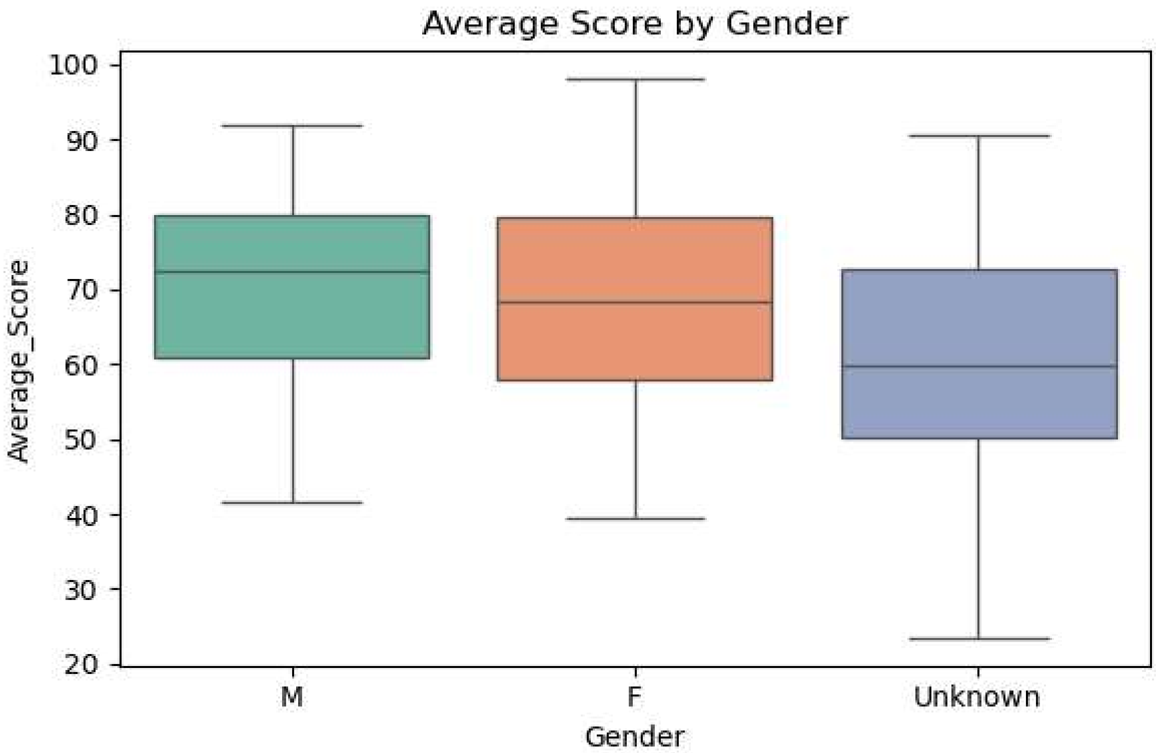
```
grade_avg = df.groupby("Grade")["Average_Score"].mean().sort_index()
plt.figure(figsize=(6, 4))
plt.bar(grade_avg.index, grade_avg.values, color="#4C72B0")
plt.title("Average Score by Grade")
plt.xlabel("Grade")
plt.ylabel("Average Score")
plt.tight_layout()
plt.show()
```



In [16]:
```
plt.figure(figsize=(6, 4))
sns.histplot(df["Average_Score"], bins=15, kde=True, color="#55A868")
plt.title("Distribution of Average Scores")
plt.xlabel("Average Score")
plt.tight_layout()
plt.show()
```

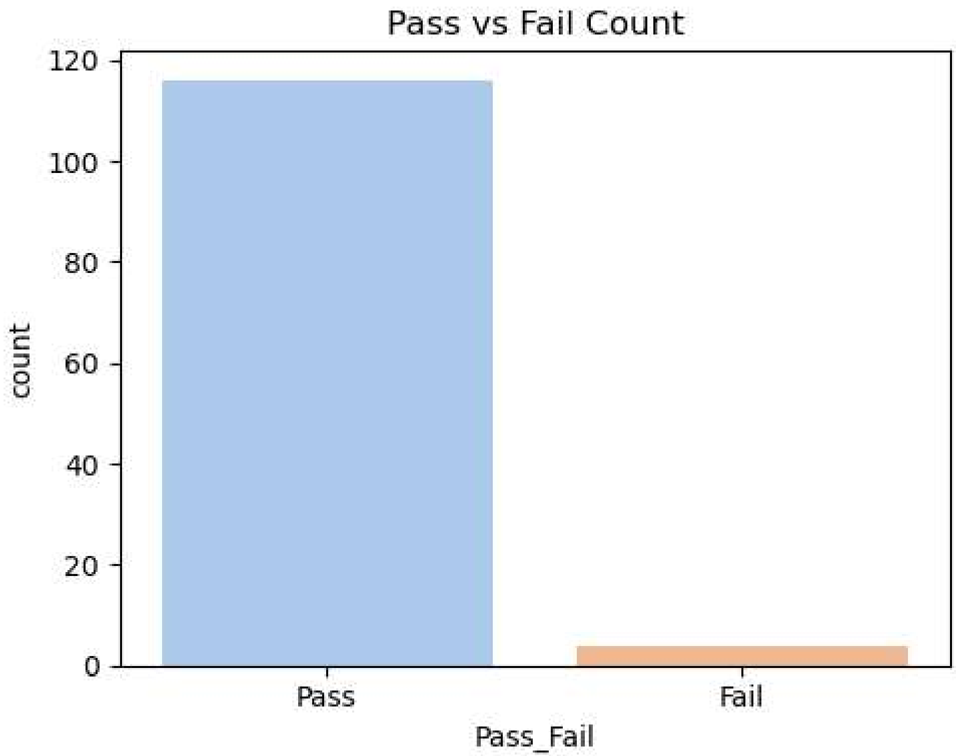


In [17]:
```
plt.figure(figsize=(6, 4))
sns.boxplot(data=df, x="Gender", y="Average_Score", hue="Gender", palette="Set2", legend=False)
plt.title("Average Score by Gender")
plt.tight_layout()
plt.show()
```



In [18]:

```
plt.figure(figsize=(5, 4))
sns.countplot(data=df, x="Pass_Fail", hue="Pass_Fail", palette="pastel", legend=False)
plt.title("Pass vs Fail Count")
plt.tight_layout()
plt.show()
```


Pass vs Fail Count
120
100
80
60
40
20
0
count
Pass
Fail
Pass_Fail

In [ ]: